Appendix I

MEDLINE Search Strategy



Search History/Alerts

Print Search History | Retrieve Searches | Retrieve Alerts | Save Searches / Alerts

Select / deselect all | Search with AND | Search with OR | Delete Searches

	Search ID#	Search Terms
☐	S4	S1 AND S2 AND S3
☐	S3	(MH "quality of life") OR "quality of life" OR "function" OR "performance" OR "skill" OR "activit*" OR "participation"
☐	S2	(MH "outcome assessment") OR "outcome" OR "assessment" OR (MH "questionnaires") OR "questionnaire" OR (MH "interviews") OR "interview" OR "measur*" OR "instrument" OR "test" OR "evaluat* OR "observ*"
☐	S1	(MH "Radius Fractures") OR "distal radius fracture" OR "distal radial fracture"

Appendix II

Search Terms

AND →

OR ↓

Participations	**Interventions**	**Outcomes**
1. Distal radius fracture 2. Distal radial fracture	4. Outcome 5. Assessment 6. Questionnaire 7. Interview 8. Measure 9. Instrument 10. Test 11. Evaluation 12. Observation	16. Quality of life 17. Function 18. Performance 19. Skill 20. Activity 21. Participation
3. Radius fracture (MeSH)	13. Outcome assessment (MeSH) 14. Questionnaires (MeSH) 15. Interviews (MeSH)	22. Quality of life (MeSH)

Appendix III

Data Extraction Table

Study Description/ Methods

Author(s)	Title	Year	Country	NHMRC Level	Therapeutic Level of Evidence	PedRO Score

Participant Characteristics

Sample Size	Age range	Age mean (SD)	Sex (male)	Sex (female)	Dominant hand (%)

Interventions

Intervention(s)	Control

Outcome measures

Time of measurement	Outcome Measure

Appendix IV. Participant and Study Characteristics of 119 Selected Studies

Authors, year	Country	National Health and Medical Research Council (NHMRC) Level of Evidence	Experimental Intervention	Control Intervention	Sample Size	Age		Sex	
						Range	Mean (SD)	Male, Number	Female, Number
(Ali et al., 2018)	Sweden	III-2 (Cohort)	Casting and percutaneous fixation	---	63	18-65	53 (9)	16	46
(Alkner et al., 2018)	Sweden	II (Randomised Controlled Trial (RCT)	Adjunctive intermittent pneumatic compression therapy following volar plating	Immobilisation only	115	50-88	63	11	104
(Andrade-Silva et al., 2018)	Brazil	II (RCT)	Splint (post-volar plate fixation)	No Splint	39	18+	49.3	17	22
(Ayong et al., 2014)	Belgium	IV (Case series)	Angle-stable volar T-plate	---	19	21-86	61.7	6	13
(Bayon-Calatayud et al., 2016)	Spain	II (RCT)	Mirror Therapy	No mirror therapy	22	26-85	58.2	7	15
(Behrendt et al., 2017)	Germany	III-2	PEEK Plate osteosynthesis	Titanium Implant	26	Not reported	59.3	5	21
(Belloti et al., 2010)	Brazil	II (RCT)	De Palma pinning	External Fixation	100	>40	58	27	73
(Bentohami et al., 2019)	Netherlands	II (RCT)	Cast immobilisation	Cast immobilisation for 4-6 weeks	72	Not reported	55.29	23	49

			shortened for 3 weeks						
(Boyer, 2015)	UK	II (RCT)	Volar Lock-plate fixation	Kirschner-wire fixation	461	Not reported	59		
(Braziulis et al., 2013)	Lithuania	IV (Case series)	Volar Locking plate fixation	---	120	Not reported	64	14	106
(Brehmer & Husband, 2014)	USA	II (RCT)	Accelerated rehabilitation protocol	Standard rehabilitation	78	Not reported	52	21	57
(Brennan et al., 2016)	Ireland	III-2 (Non-randomised, experimental trial)	Volar Plate	K-wire fixation	318	20-65	46.7	112	206
(Bruder et al., 2016)	Australia	II (RCT)	Progressive exercise and structured advice program	Structured advice program	33	21+	54 (18)	8	25
(Chen et al., 2017)	China	II (RCT)	3D Printing technology for preoperative planning	Routine Approach	107	Not reported	28 (6)	65	42
(Cho et al., 2014)	Korea	III-2 (Cohort)	AO LDRS Volar Plates	Acu-Loc Volar Plates	52	28-79	57.9	17	35
(Costa et al., 2019)	UK	II (RCT)	K-wire fixation (Kirschner)	Locking plates	448	19-89	58	37	194
(D'Agostino & Barbier, 2013)	Belgium	II (RCT)	AlloMatrix bone graft	No graft	50	17-69	43	21	29
(Dilek et al., 2018)	Spain	II (RCT)	Graded motor imagery and traditional rehabilitation	Traditional rehabilitation	36	18-65	50	12	24
(Durama z et al., 2018)	Turkey	III-3 (Retrospective cohort)	Volar Locking Plate	K-wire	114	18-86	44.9 (15.4)	71	42
(Farías Cisneros et al., 2010)	Mexico	III-2 (Non-randomised, experimental trial)	Closed reduction (CR) and immobilisation with a splint or cast, CR and internal fixation (IF) with percutaneous nails and immobilisation with a splint or cast, CR and external fixation (EF) either alone or combined with percutaneous nails, open education OR and IF with volar or dorsal plates	---	109	15-90	59.8	29	80
(Filipova et al., 2015)	Slovenia	II (RCT)	Physical and Occupational therapy	Physical therapy	61	31-84	60	14	47
(Gereli et al.,	Turkey	III-2 (Non-randomised	Closed reduction cannulate screw	Open reduction palmar locking	65	18-60	43	33	32

2014)		, experimental trial)	fixation	plate fixation					
(Gradl et al., 2016)	Germany	II (RCT)	Volar Lock-plate fixation	Intramedullary nailing	28	Not reported	64	4	24
(Gradl et al., 2013)	Germany	II (RCT)	Plate osteosynthesis	Non-bridging external fixation	102	18-88	63	13	89
(Gradl et al., 2014)	Germany	II (RCT)	Volar Locking Plate fixation	Intramedullarly nailing	121		62	18	103
(Grafstein et al., 2010)	USA	II (RCT)	Circumferential Casting	volar-dorsal splint & modified sugar-tong splinting	101	Not reported	58	21	80
(Grewal et al., 2011)	Canada	II (RCT)	Open reduction and internal fixation	External Fixation	50	Not reported	55	12	38
(Grle et al., 2017)	Bosnia and Herzegovina	II (Prospective Cohort)	Dorsal forearm splint with wrist in dorsal flexion	Dorsal forearm splint with wrist in palmar flexion	100	Not reported	63	25	75
(Gruber et al., 2010)	Austria	IV (Case series)	Volar Plate Fixation	---	54	24-93	63	17	37
(Hakimi, Jungbluth, Gehrmann, et al., 2010)	Germany	III-2 (Non-randomised, experimental trial)	Unilateral palmar angularly stable plate osteosynthesis	Multidirectional palmar angularly stable plate osteosynthesis	40	25-95	61	9	31
(Hakimi, Jungbluth, Windolf, et al., 2010)	Germany	IV (Case series)	Locking palmar plate	---	89	18-94	62	30	59
(Hammer et al., 2019)	Norway	II (RCT)	Volar Locking Plate	Augmented External Fixation	166	18-70	55	52	114
(Han et al., 2015)	China	III-2 (Non-randomised, experimental trial)	External fixator with T-plate internal fixation	Palmar T-plate fixation	61	25-68	48.5	27	34
(Hill et al., 2018)	USA	II (RCT)	Immobilisation in maximal supination with a sugar-tong splint	Immobilisation in no restriction of supination with a volar splint	32	Not reported	43	19	13
(Hove et al., 2010)	Norway	II (RCT)	Dynamic fixator	Static Fixator	70	Not reported	54 (6)	19	51
(Huh et al., 2012)	Korea	IV (Case series)	Open Reduction Internal Fixation	---	34	21-74	55	14	20
(Hull et al., 2011)	USA	III-2 (Non-randomised, experimental trial)	Volar Locking Plate fixation	K-wire fixation	71	18-86	61	16	55
(Iselin et al., 2016)	Switzerland	IV (Case series)	Combined Palmar and Dorsal Internal Fixation	---	12	24-73	47	12	0

(Javed et al., 2015)	UK	IV (Case series)	Volar Locking plate fixation	---	62	17-86	52.5	22	40
(Jeudy et al., 2012)	France	II (RCT)	External fixation	Open Reduction Internal Fixation with locked volar plating	75	Not reported	64	18	57
(Jongs et al., 2012)	Australia	II (RCT)	Dynamic splint	No splint	40	52-72	63	12	28
(Kamal & Ruch, 2017)	USA	IV (Case series)	Volar capsule release	---	11	21-62	45	5	6
(Karantana et al., 2013)	UK	II (RCT)	Volar Locking Plate	Conventional Percutaneous	130	18-73			
(Khan et al., 2017)	Pakistan	II (RCT)	K-wire fixation (Kirschner)	T-plate	30	25-55	40	18	12
(Khan et al., 2016)	Pakistan	IV (Case series)	Volar Lock-plate fixation	---	43	20-95	47 (14)	27	16
(Kim et al., 2014)	South Korea	IV (Case series)	Brachioradialis release	Brachioradialis preserved	42	20-70	58	7	35
(Kim & Park, 2013)	Korea	III-2 (Case-control)	Volar Plate Fixation for open fracture	Volar plate fixation for closed DRF	58	18+	53	23	35
(Kumaravel & Karthikeyan, 2015)	India	II (RCT)	cast immobilization	External Fixation	48	38-70	50	17	31
(Kumbaraci et al., 2014)	Turkey	III-2 (Cohort)	External fixation	Volar locking plates	80	69	50	35	34
(Kuo et al., 2013)	Taiwan	II (RCT)	Early digit mobilisation	Conventional treatment	22	Not reported	62	7	15
(Landgren et al., 2017)	Sweden	II (RCT)	Fragment-Specific Fixation	Volar locking plates	50	21-69	56	11	39
(Landgren et al., 2011)	Sweden	II (RCT)	External fixation	Internal Fixation	50	20-65	53	14	37
(Lattmann et al., 2011)	Switzerland	IV (Case series)	Volar locking plate osteosynthesis (VLPO)	---	245	18-96	62 (18)	60	185
(Lauder et al., 2015)	USA	IV (Case series)	Dorsal bridge plate fixation	---	18	29-89	61 (4)	13	5
(Loisel et al., 2015)	France	III-2 (Non-randomised, experimental trial)	Volar Locking Plate	Percutaneous Pinning	52	43-82	63.6	5	47
(Ma et al., 2017)	China	II (RCT)	Brachioradialis release	No brachioradialis release	74	Not reported	50 (15)	38	36
(Magnus et al., 2013)	Canada	II (RCT)	Standard rehabilitation plus strength training	Standard rehabilitation (forearm casting and hand exercises)	51	50+	63	0	39

(Mahmoud et al., 2012)	Egypt	IV (Case series)	Corrective osteomy and a Volar Locking Plate without bone graft	---	19	20-49	31.8	17	2
(Manrique et al., 2017)	Colombia	II (RCT)	Percutaneous pinning	Internal Fixation	67	19-84	53	16	51
(Marcano et al., 2015)	USA	IV (Case series)	Operative reduction and fracture fixation using a volar buttress plate	---	253	19-87	52	120	162
(Matschke et al., 2011)	Germany	IV (Case series)	Angle-stable volar T-plate	---	118	20-80	57 (15.2)	36	82
(Mirhamidi & Bayat, 2013)	Iran	II (RCT)	Extra-focal technique	Kapandji technique	45	17-84	45	25	20
(Mishra et al., 2018)	India	II (RCT)	External fixation	Open reduction with volar plate surgery	80	18-60	37	63	17
(Mitsukane et al., 2015)	Japan	II (RCT)	Repetitive Wrist Extension exercises	No exercises	28	27-81	63 (13)	9	19
(Mulders et al., 2019)	Netherlands	II (RCT)	Open reduction and volar plate fixation	Plaster immobilisation	90	18-75	59	23	67
(Namazi & Mehbudi, 2016)	Iran	II (RCT)	Closed reduction and percutaneous pinning with Platelet-rich Plasma	Closed reduction and percutaneous pinning without Platelet-rich Plasma	30	18-50	32	24	5
(Neidenbach et al., 2010)	UK	III-2 (Case-control)	Conservatively without closed reduction	Conservatively with closed reduction	83		62	7	76
(Opel et al., 2014)	London	IV (Case series)	Corrective osteomy and a fixed-angle volar Locking Plate	---	20	19-83	57	4	16
(Osti et al., 2012)	Austria	III-3 (Retrospective Cohort)	Titanium nonlocking T-plates	Titanium locking plates	60	Not reported	52	18	42
(Ozer et al., 2011)	USA	III-2 (Case-control)	Allograft bone chips following volar plating	No bone graft with volar plating	28	15-65	35	17	12
(Perugia et al., 2017)	Italy	III-2 (Case-control)	CarboFix volar locking plate	Acu-Loc Volar Distal Radius Plate	30	32-71	54	9	21
(Phadnis et al., 2012)	UK	IV (Case series)	Internal fixation with the Stryker Matrix volar locking plate	---	183	16-93	62.4	51	132
(Pillukat et al., 2013)	Germany	IV (Case series)	Reconstruction with subchondral K-wires and bone grafts	---	6	18-67	39	5	1
(Plate et al., 2015)	USA	II (RCT)	Intramedullary nails	Volar locking plates	60	18-80	55 (15)	16	44
(Quadlbauer et al., 2017)	Austria	II (RCT)	Early immobilisation	Immobilisation group	28	Not reported	54	4	24

(Rahman et al., 2012)	Pakistan	II (RCT)	Above elbow Plaster of Paris cast	External Fixation with distraction	60	32-58	42.7 (7)	22	38
(Ratajczak & Płomiński, 2015)	Poland	II (RCT)	Isometric massage, physical therapy and exercise	No isometric massage	40	25-60	42	18	22
(Roh et al., 2015)	Korea	II (RCT)	Volar Locking Plate	External fixation	74	Not reported	54 (11)	Not reported	Not reported
(Ruckenstuhl et al., 2014)	Austria	IV (Case series)	Volar locking plate	---	39	28-87	61 (8)	15	24
(Saddiki et al., 2012)	France	II (RCT)	Py's (isoelastic pinning) technique	Kapandji's (intrafocal pinning) techniques	97	18+	63	14	83
(Saremi et al., 2016)	Iran	III-2 (Cohort)	Posterior Interosseous Bone Flap	---	9	34-78	55		
(Saving et al., 2019)	Sweden	II (RCT)	External fixation	Volar locking plates	118	50-74	63	9	109
(Schneiders et al., 2012)	Germany	IV (Case series)	K-wire osteosynthesis	---	100	Not reported	62	33	67
(Schønnemann et al., 2011)	Denmark	II (RCT)	Non-bridging external fixation	Intramedullary fixation	61	19-88	62	10	51
(Selles et al., 2019)	Netherlands	II (RCT)	Open Reduction Internal Fixation with arthroscopic debridement	Open Reduction Internal Fixation without arthroscopic debridement	50	26-66	59	17	33
(Sharma et al., 2014)	India	II (RCT)	Volar locking plate (open reduction and fixation)	Nonoperative methods	64	25-55	50	26	38
(Shetty et al., 2011)	India	IV (Case series)	Open Reduction Internal Fixation with volar locking plate	---	23	21-74	35	14	9
(Shukla et al., 2014)	India	II (RCT)	External fixation	Volar locking plates	110	>18	39 (13)	49	61
(Sonntag et al., 2019)	Denmark	II (RCT)	Repair of pronator quadratus muscle	No repair of PQ muscle	59	>18	61	15	44
(Souer et al., 2010)	USA	III-2 (Case-control)	Titanium locked plate	Stainless-steel oblique plate	62	Not reported	58	18	44
(Stuby et al., 2015)	Germany	II (RCT)	Dorsal plaster splint	Vacuum-fit flexible but blocked orthosis	29	20-75	48	44	13
(Szekeres et al., 2018)	Canada	II (RCT)	Moist hot pack	Therapeutic whirlpool bath	60	Not reported	53	11	49
(Tanaka et al., 2016)	Japan	II (RCT)	AcuLoc distal-type fixation plate	VariAx proximal-type fixation plate	64	>20	62 (14)	15	49
(Tang et al.,	China	III-2 (Non-randomised	Volar anatomical plates	Locking plates	33	22-72	48	19	14

2012)		, experimental trial)							
(Tarng et al., 2011)	Taiwan	IV (Case series)	Palmar locking plates	---	7	34-76	55	3	4
(Toon et al., 2017)	Singapore	III-2 (Non-randomised, experimental trial)	Open Reduction Internal Fixation	Nonoperative methods	60	23-79	55	25	35
(Twigt et al., 2013)	Netherlands	IV (Case series)	Spring plate technique	---	34	18-80	49	11	23
(Valentini et al., 2014)	Italy	III-2 (Non-randomised, experimental trial)	Fixed-hole plates	Variable-hole plates	52	22-82	58	16	36
(van Gerven et al., 2019)	Netherlands	II (RCT)	Usual care	Reduced Imaging	326	18+	56 (18)	78	248
(van Leerdam et al., 2019)	Netherlands	III-2 (Non-randomised, experimental trial)	Operative	Nonoperative methods	272	18+	62 (16)	69	203
(Vargaonkar, 2014)	India	III-2 (Non-randomised, experimental trial)	Open reduction and internal fixation with a volar plate	Closed reduction and Plaster of Paris cast, or Distractor application	60	20-70	46	25	35
(Venkatesh et al., 2016)	India	II (RCT)	Closed reduction and cast application	Percutaneous K-wire fixation	60	24-70	47		
(Vlček et al., 2014)	Czech Republic	III-2 (Non-randomised, experimental trial)	Intramedullary X-screw osteosynthesis	Open Reduction Internal Fixation with locked volar plating with variable-angle screw insertion	68	22-77	55	17	51
(Wada et al., 2011)	Japan	III-2 (Non-randomised, experimental trial)	Opening-wedge osteomy	Closing-wedge osteomy	20	19-79	58	4	16
(Wadsten et al., 2018)	Sweden	III-2 (Cohort)	Conservatively treated	---	175	16-75	55	40	135
(Watson et al., 2018)	Australia	II (RCT)	Open Reduction Internal Fixation for 1 weeks	Open Reduction Internal Fixation for 3 or 6 weeks	133	18+	53 (15)	48	85
(Wei et al., 2016)	China	IV (Case series)	Minimally invasive percutaneous plate osteosynthesis (MIPPO)	---	9	38-71	55 (10)	4	5
(Wijffels et al., 2012)	USA	IV (Case series)	Extended FCR approach and volar locked plate	---	35	22-82	51	10	25

			and screw fixation						
(Wilcke et al., 2011)	Sweden	II (RCT)	Volar Locking Plate	External fixation	63	20-69	55	15	48
(Wollstein et al., 2019)	USA	II (RCT)	Standard postoperative therapy and sensorimotor postoperative therapy	Standard postoperative therapy	66	Not reported	63 (18)	20	46
(Yamazaki et al., 2015)	Japan	II (RCT)	Fluoroscopic guided reduction	Arthroscopic guided reduction	74	24-92	65	20	54
(Yang et al., 2018)	Singapore	IV (Case series)	Conservatively treated	---	138	Not reported	59 (16)	45	93
(Ydreborg et al., 2015)	Sweden	IV (Case series)	Plate fixation	---	101	18+	58 (13)	19	82
(Yin et al., 2017)	China	IV (Case series)	Arthroscopically assisted K-wire with external fixation	---	21	32-57	42	14	7
(Yu et al., 2019)	China	IV (Case series)	External fixation	Volar locking plate fixation	62	18+	51 (15)	36	26
(Zhang et al., 2016)	China	III-2 (Non-randomised, experimental trial)	Volar Locking Plate	Non-locking plate	57	18-60	39 (13)	37	30
(Zhang et al., 2017)	China	II (RCT)	Volar locking plate with 1.5-2cm longitudinal incision	Volar lock plate with conventional flexi-carpi radialis approach	157	18-67	42	95	62
(Zelai et al., 2018)	China	II (RCT)	Traditional Henry approach incision reduction oblique T-lock steel plate internal fixation	Minimally invasive percutaneous palm implantation of oblique T-shaped locking plate fixation	56	30-67	46	33	23
(Zyluk et al., 2018)	Poland	III-2 (Non-randomised, experimental trial)	K-wire fixation (Kirschner)	Palmar plate fixation	102	36-87	54	23	79

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